

Catherine Hou

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EDUCATION

Wesleyan University, Middletown, CT

May 2026

Bachelor of Arts, GPA: 3.90/4.00 (Major GPA 3.84)

Double Major: Astronomy & Psychology

Related Coursework: Astrogeochemistry • Introduction to Astrophysics • Observational Astronomy • Waves and Oscillations • E&M • Calc I&II • Vectors & Matrices • Galaxies and Quasars • Nuclear Astrophysics • Galactic Structure and Formation •

PUBLICATIONS (3):

- ARKS II: The Radial Structure of Debris Discs (co-author) **June 2025**
- ARKS III: The Vertical Structure of Debris Discs (co-author) **June 2025**
- [She's a New Yorker](#) under Ms. Magazine **March 2021**

EXPERIENCE

REU Intern, **American Museum of Natural History**, New York City, NY

May 2025 - August 2025

- Developed a Django-based tool for the Strong Lensing Database (SLED), enabling researchers to upload and share lens models.
- Improved findability, accessibility, interoperability, and reproducibility (FAIR principles) of lens modeling data.
- Contributed to and interpreted SLED's large and complex codebase, leveraging GitHub for version control and integrating Python scripts to enhance functionality.
- Enhanced science communication and public speaking skills by participating in daily workshops and presenting a 10-minute talk at the AMNH REU Summer Symposium. Improved public speaking, science outreach, and presentation skills.

Research Fellow, **Wesleyan University**, Middletown, CT

June 2024 - August 2024

- Worked on the Alma Survey to Resolve exoKuiper belt Substructures (ARKS) to parametrically model debris disk substructures using Python, CANFAR, CASA, and other coding interfaces. Improved computational proficiency.
- Ran MCMC chains, graphed Fourier transforms from DMR plots in Python to analyze irregular residual data, and systematized data from literature to calculate dynamical masses of debris disks, supporting comparative astrophysical analysis.
- Created a summative poster accessible to all scientific backgrounds and presented work at KNAC, enhancing public speaking confidence, learning presentation aesthetics, and strengthening individualized analytical thinking patterns.

Summer Associate, **Civic Changemakers**, Washington, DC

June 2023 - August 2023

- Worked with the Emerson Collective (ECYC) and AmeriCorps to develop a curriculum on civic engagement and politics. Taught students from disenfranchised communities in DC how to build self-sufficient advocacy networks.
- Developed and oversaw a project addressing youth homelessness, raising 650 pounds of food donations and distributing them to 3 different youth shelters in Wards 7 and 8 of DC.
- Authored and introduced program policies on restorative justice and cultural sensitivity, later adopted by the ECYC program. Developed new lesson plans focused on communicating across differences and addressing ingrained biases.

LEADERSHIP AND TEACHING

Community Advisor and House Manager for **Residential Life**, Wesleyan University

August 2024 - Present

- Improved interpersonal and organizational skills by organizing community events and filing communication reports.

Tour Guide for **The Admissions Office**, Wesleyan University

October 2023-Present

- Developed leadership, teamwork, and public speaking skills while providing networking growth.

Teaching Assistant for **Introduction to Astrophysics**, Wesleyan University

August 2024-December 2024

- Improved teaching, time management, and adaptability skills. Ran problem-solving sessions and enhanced organizational skills.

SKILLS AND INTERESTS

Languages: English [Fluent], Spanish [Intermediate – conversational/proficient]

Programming and Data Science: Python (NumPy, Pandas, Matplotlib), Django, CASA, CANFAR, SPSS, UNIX, LaTeX

Arts: Piano (Jazz improv, classical, contemporary), Guitar, Percussion (Drums, mallets, ensemble), Performing Arts (Magic)